

CO3_AT1:CASE STUDY

Question

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why.

Step 1: Arrange the Data

The data is already arranged in ascending order:

4, 5, 5, 6, 7, 8, 8, 9, 30

Step 2: Calculate the Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

Where,

ΣX = Sum of all observations

n = Number of observations

Step 2.1: Find the sum.

$$\begin{aligned}\Sigma X &= 4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 \\ &= 82\end{aligned}$$

Step 2.2: Count the observations.

$$n = 9$$

Step 2.3: Substitute into the formula.

$$\text{Mean} = 82/9$$

$$= 9.11$$

Therefore, the Mean Salary = 9.11 lakhs.

Step 3: Calculate the Median

Formula:

Median is the middle observation after arranging the data.

Since $n = 9$ (an odd number),

$$\begin{aligned}\text{Middle Position} &= (n + 1)/2 \\ &= (9 + 1)/2 \\ &= 5\text{th position}\end{aligned}$$

The 5th value is 7.

Therefore,

Median = 7 lakhs.

Step 4: Calculate the Mode

Mode is the value that occurs most frequently.

Frequency Table

Value	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

The highest frequency is 2.

Both 5 and 8 occur twice.

Therefore,

Mode = 5 and 8 (Bimodal Distribution).

Step 5: Calculate the Range

Formula:

Range = Highest Value – Lowest Value

Highest Salary = 30

Lowest Salary = 4

$$\begin{aligned}\text{Range} &= 30 - 4 \\ &= 26\end{aligned}$$

Therefore, Range = 26 lakhs.

Step 6: Variance and Standard Deviation

x	x – Mean	(x – Mean) ²	Rounded
4	-5.1111	26.123457	26.12
5	-4.1111	16.901235	16.90
5	-4.1111	16.901235	16.90
6	-3.1111	9.679012	9.68
7	-2.1111	4.456790	4.46
8	-1.1111	1.234568	1.23
8	-1.1111	1.234568	1.23
9	-0.1111	0.012346	0.01
30	20.8889	436.345679	436.35

Step 6.1: Add all squared deviations.

$$\Sigma(x - \text{Mean})^2 = 512.888889$$

Step 6.2: Variance

Formula:

$$\text{Variance} = \Sigma(x - \text{Mean})^2 / n$$

$$\begin{aligned}\text{Variance} &= 512.888889/9 \\ &= 56.987654\end{aligned}$$

Variance ≈ 56.99

Step 6.3: Standard Deviation

Formula:

Standard Deviation = $\sqrt{\text{Variance}}$

= $\sqrt{56.987654}$

= 7.549017

≈ 7.55

Step 7: Identify the Outlier

Among all salaries, 30 lakhs is much larger than every other value (which lies between 4 and 9 lakhs).

Hence,

Outlier = 30 lakhs.

Step 8: Best Measure of Central Tendency

The three measures are:

Mean = 9.11

Median = 7

Mode = 5 and 8

The mean is increased because of the extremely high salary of 30 lakhs.

The median depends only on the middle value and is not greatly affected by the outlier.

Therefore, the median is the best measure to represent the typical employee salary.

Final Answers

Mean = 9.11 lakhs

Median = 7 lakhs

Mode = 5 and 8

Range = 26 lakhs
Variance = 56.99
Standard Deviation = 7.55
Outlier = 30 lakhs
Best Measure = Median

2.Question

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

Step 1: Arrange the Data

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Step 2: Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

$$\begin{aligned}\Sigma X &= 120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 \\ &= 1505\end{aligned}$$

$$n = 10$$

$$\begin{aligned}\text{Mean} &= 1505/10 \\ &= 150.50 \text{ ms}\end{aligned}$$

Step 3: Median

Since there are 10 observations (an even number), the median is the average of the 5th and 6th observations.

5th observation = 132

6th observation = 135

$$\begin{aligned}\text{Median} &= (132 + 135) / 2 \\ &= 267 / 2 \\ &= 133.5 \text{ ms}\end{aligned}$$

Step 4: Range

Formula:

$$\text{Range} = \text{Highest Value} - \text{Lowest Value}$$

$$\text{Highest} = 300$$

$$\text{Lowest} = 120$$

$$\begin{aligned}\text{Range} &= 300 - 120 \\ &= 180 \text{ ms}\end{aligned}$$

Step 5: Variance and Standard Deviation

x	x-Mean	(x-Mean) ²	Rounded
120	-30.50	930.25	930.25
125	-25.50	650.25	650.25
128	-22.50	506.25	506.25
130	-20.50	420.25	420.25
132	-18.50	342.25	342.25
135	-15.50	240.25	240.25
140	-10.50	110.25	110.25
145	-5.50	30.25	30.25
150	-0.50	0.25	0.25
300	149.50	22350.25	22350.25

Step 5.1:

$$\Sigma(x-\text{Mean})^2 = 25580.50$$

Step 5.2:

$$\begin{aligned}\text{Variance} &= \Sigma(x - \text{Mean})^2 / n \\ &= 25580.50 / 10 \\ &= 2558.05\end{aligned}$$

Step 5.3:

$$\begin{aligned}\text{Standard Deviation} &= \sqrt{\text{Variance}} \\ &= \sqrt{2558.05} \\ &= 50.58 \text{ ms}\end{aligned}$$

Step 6: Identify the Skewness

The response time of 300 ms is much larger than the remaining values. This creates a long tail towards the higher values.

Therefore, the data is positively skewed (right-skewed).

Step 7: Why is the Median More Useful?

Mean = 150.50 ms

Median = 133.5 ms

The response time of 300 ms is an outlier. It increases the mean considerably, making the average response time appear slower than usual.

The median depends only on the middle observations and is not greatly affected by this extreme value. Therefore, the median provides a better representation of the typical website response time.

Final Answers

Mean = 150.50 ms

Median = 133.5 ms

Range = 180 ms

Variance = 2558.05

Standard Deviation = 50.58 ms

Distribution = Right-skewed

Best Measure = Median

3.Question

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the IQR.
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

Step 1: Arrange the Data

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Step 2: Find Q2 (Median)

There are 12 observations (an even number).

Median = Average of the 6th and 7th observations.

6th observation = 52

7th observation = 55

$$\begin{aligned} Q2 &= (52 + 55) / 2 \\ &= 53.5 \end{aligned}$$

Step 3: Find Q1

First divide the data into two equal halves.

Lower Half:

42, 45, 48, 50, 52, 52

Q1 is the median of the lower half.

$$\begin{aligned} Q1 &= (48 + 50) / 2 \\ &= (48 + 50) / 2 \\ &= 49.0 \end{aligned}$$

Step 4: Find Q3

Upper Half:

55, 58, 60, 62, 65, 95

Q3 is the median of the upper half.

$$\begin{aligned} Q3 &= (60 + 62) / 2 \\ &= (60 + 62) / 2 \\ &= 61.0 \end{aligned}$$

Step 5: Calculate the IQR

Formula:

$$IQR = Q3 - Q1$$

$$\begin{aligned} IQR &= 61.0 - 49.0 \\ &= 12.0 \end{aligned}$$

Step 6: Apply the $1.5 \times IQR$ Rule

Step 6.1:

$$\begin{aligned} 1.5 \times IQR &= 1.5 \times 12.0 \\ &= 18.0 \end{aligned}$$

Step 6.2:

$$\begin{aligned} \text{Lower Fence} &= Q1 - (1.5 \times IQR) \\ &= 49.0 - 18.0 \\ &= 31.0 \end{aligned}$$

$$\begin{aligned} \text{Upper Fence} &= Q3 + (1.5 \times IQR) \\ &= 61.0 + 18.0 \\ &= 79.0 \end{aligned}$$

Any value below 31.0 or above 79.0 is an outlier.

Therefore, Outlier = 95.

Step 7: Calculate the Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

$$\begin{aligned}\Sigma X &= 42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 \\ &= 684\end{aligned}$$

$$\begin{aligned}\text{Mean} &= 684/12 \\ &= 57.00\end{aligned}$$

Step 8: Compare Mean and Median

$$\text{Mean} = 57.00$$

$$\text{Median (Q2)} = 53.5$$

The score 95 is much larger than the other marks. It pulls the mean upward, making the average higher than the performance of most students.

The median depends only on the middle values and therefore changes very little.

Hence, the outlier affects the mean much more than the median.

Final Answers

$$Q1 = 49.0$$

$$Q2 = 53.5$$

$$Q3 = 61.0$$

$$\text{IQR} = 12.0$$

$$\text{Outlier} = 95$$

$$\text{Mean} = 57.00$$

$$\text{Median} = 53.5$$

Conclusion:

The value 95 is an outlier because it lies above the upper fence (79.0). It increases the mean but has very little effect on the median.

4.Question

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.

2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Step 1: Arrange the Data

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Step 2: Calculate the Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

$$\begin{aligned}\Sigma X &= 18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 \\ &= 250\end{aligned}$$

$$n = 10$$

$$\begin{aligned}\text{Mean} &= 250/10 \\ &= 25.00 \text{ orders}\end{aligned}$$

Step 3: Calculate the Median

Since there are 10 observations (an even number), the median is the average of the 5th and 6th observations.

$$\begin{aligned}\text{Median} &= (24 + 25) / 2 \\ &= 49 / 2 \\ &= 24.5\end{aligned}$$

Step 4: Calculate the Mode

Frequency of each value:

18 → 1

20 → 1

22 → 2

24 → 1

25 → 1

26 → 1

28 → 1

30 → 1

35 → 1

The value 22 occurs most frequently.

Mode = 22

Step 5: Variance and Standard Deviation

x	x-Mean	(x-Mean) ²	Rounded
18	-7.00	49.00	49.00
20	-5.00	25.00	25.00
22	-3.00	9.00	9.00
22	-3.00	9.00	9.00
24	-1.00	1.00	1.00
25	0.00	0.00	0.00
26	1.00	1.00	1.00
28	3.00	9.00	9.00
30	5.00	25.00	25.00
35	10.00	100.00	100.00

$$\Sigma(x-\text{Mean})^2 = 228.00$$

$$\text{Variance} = \Sigma(x-\text{Mean})^2 / n$$

$$= 228.00/10$$

$$= 22.80$$

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

$$= \sqrt{22.80}$$

$$= 4.77 \text{ orders}$$

Step 6: Coefficient of Variation (CV)

Formula:

$$\text{CV} = (\text{Standard Deviation} / \text{Mean}) \times 100$$

$$\begin{aligned} CV &= (4.77 / 25.00) \times 100 \\ &= 19.10\% \end{aligned}$$

Step 7: Interpretation of Standard Deviation

The standard deviation is 4.77 orders.

This means that the daily number of orders typically differs from the average (25.00 orders) by about 4.77 orders.

A smaller standard deviation indicates more consistent daily orders, while a larger one indicates greater fluctuations.

Step 8: Is the Data Highly or Moderately Variable?

The coefficient of variation is 19.10%.

Since the CV is below 20%, the variation is considered moderate. The order data is fairly stable with only moderate day-to-day changes.

Final Answers

Mean = 25.00

Median = 24.5

Mode = 22

Variance = 22.80

Standard Deviation = 4.77

Coefficient of Variation = 19.10%

Conclusion:

The company experiences moderate variation in daily orders. The standard deviation shows that daily orders stay reasonably close to the average, helping the company plan inventory and staffing more effectively.

5.Question

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.

2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

MODEL A

Data: 2, 3, 3, 4, 4, 5, 5, 6

Step 1: Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

$$\begin{aligned}\Sigma X &= 2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 \\ &= 32\end{aligned}$$

$$n = 8$$

$$\begin{aligned}\text{Mean} &= 32/8 \\ &= 4.00\end{aligned}$$

x	x-Mean	(x-Mean) ²	Rounded
2	-2.00	4.0000	4.00
3	-1.00	1.0000	1.00
3	-1.00	1.0000	1.00
4	0.00	0.0000	0.00
4	0.00	0.0000	0.00
5	1.00	1.0000	1.00
5	1.00	1.0000	1.00
6	2.00	4.0000	4.00

Step 2: Variance

$$\Sigma(x-\text{Mean})^2 = 12.0000$$

$$\begin{aligned}\text{Variance} &= \Sigma(x-\text{Mean})^2 / n \\ &= 12.0000/8 \\ &= 1.5000\end{aligned}$$

Step 3: Standard Deviation

$$\begin{aligned}\text{Standard Deviation} &= \sqrt{\text{Variance}} \\ &= \sqrt{1.5000} \\ &= 1.2247\end{aligned}$$

MODEL B

Data: 1, 1, 2, 4, 5, 7, 8, 9

Step 1: Mean

Formula:

$$\text{Mean} = \Sigma X / n$$

$$\begin{aligned}\Sigma X &= 1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 \\ &= 37\end{aligned}$$

$$n = 8$$

$$\begin{aligned}\text{Mean} &= 37/8 \\ &= 4.62\end{aligned}$$

x	x-Mean	(x-Mean) ²	Rounded
1	-3.62	13.1406	13.14
1	-3.62	13.1406	13.14
2	-2.62	6.8906	6.89
4	-0.62	0.3906	0.39
5	0.38	0.1406	0.14
7	2.38	5.6406	5.64

8	3.38	11.3906	11.39
9	4.38	19.1406	19.14

Step 2: Variance

$$\Sigma(x - \text{Mean})^2 = 69.8750$$

$$\begin{aligned}\text{Variance} &= \Sigma(x - \text{Mean})^2 / n \\ &= 69.8750 / 8 \\ &= 8.7344\end{aligned}$$

Step 3: Standard Deviation

$$\begin{aligned}\text{Standard Deviation} &= \sqrt{\text{Variance}} \\ &= \sqrt{8.7344} \\ &= 2.9554\end{aligned}$$

Step 4: Compare the Two Models

Model A

Mean Error = 4.00

Variance = 1.50

Standard Deviation = 1.22

Model B

Mean Error = 4.62

Variance = 8.73

Standard Deviation = 2.96

Interpretation:

- Model A has a much smaller variance and standard deviation.
- The prediction errors are clustered close to the mean.
- Model B has a much larger spread in its errors, indicating greater inconsistency.

Therefore, Model A gives more consistent predictions.

Step 5: Can the Model with the Lower Mean Error Always Be Considered Better?

No.

A lower mean error alone does not guarantee that a model is better.

A good machine-learning model should also be evaluated using:

- Variance (consistency of errors)
- Standard deviation (spread of errors)
- Presence of large prediction errors (outliers)
- Performance on unseen or test data
- Accuracy and reliability for the intended application

For example, a model may have a slightly lower mean error but produce very large errors for some predictions. Such a model may be less reliable than a model whose errors are consistently small.

Final Answers

Model A

Mean Error = 4.00

Variance = 1.50

Standard Deviation = 1.22

Model B

Mean Error = 4.62

Variance = 8.73

Standard Deviation = 2.96

More Consistent Model: Model A

Conclusion:

Model A is preferred because it produces more consistent prediction errors. However, when selecting a machine-learning model, consistency, robustness, and performance on new data should all be considered rather than relying only on the mean error.